

# HARISH KUMAR SAMALA

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## PROFESSIONAL SUMMARY

Software Engineer with 3+ years of Information Technology experience delivering end-to-end system development for complex technology platforms. Skilled in Python, SQL, and Machine Learning model development, with hands-on experience building Large Language Model (LLM) and Natural Language Processing (NLP) applications, including extracting and analyzing data from healthcare documents. Experienced in data engineering with PySpark and cloud platforms (AWS, Azure), building and validating statistical and machine learning models, performing system testing and defect resolution, and collaborating with cross-functional teams to define application scope, technical specifications, and business requirements.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, SQL

**Generative AI & LLMs:** Large Language Models, Retrieval-Augmented Generation, Agentic AI, Multi-Agent Systems, Prompt Engineering, Semantic Search, Embedding Models, OpenAI API, Claude, Gemini, Mistral, Groq, LLaMA 3, Hugging Face, LoRA, QLoRA, PEFT, Fine-Tuning, Instruction Tuning.

**AI Frameworks & Libraries:** LangChain, LangGraph, AutoGen, CrewAI, Scikit-learn, Pandas, NumPy, SciPy, PySpark, Ray, BentoML

**Databases:** PostgreSQL, MySQL, MongoDB

**Vector Databases & Search:** Pinecone, FAISS, Weaviate, ChromaDB, Hybrid Search, Vector Indexing

**Data & ML:** Pandas, NumPy, PySpark, Scikit-learn, ETL Pipelines, Sentiment Analysis, Data Visualization

**Cloud & DevOps:** AWS, Azure, Docker, CI/CD, Jenkins, Azure DevOps, GitHub, Git

**Testing & Tools:** Nginx, JMeter, dotenv

## PROFESSIONAL EXPERIENCE

### AI/ML Software Engineer

Jul 2024 – Present

#### Advanced Knowledge Technologies, NJ, USA

- Developed production Conversational RAG systems using LangChain, Groq, FAISS/ChromaDB, and Streamlit, reducing support response time by 40% and improving retrieval precision by 35%.
- Performed system testing, defect resolution, and validation across AI/ML applications and backend services.
- Built and validated machine learning models using Scikit-learn, Apriori Algorithm, and Sentiment Analysis.
- Designed ETL pipelines using PySpark and Pandas for document ingestion, embedding generation, and semantic search.
- Developed RESTful APIs using Flask and Django and integrated CI/CD pipelines with Azure DevOps.
- Built reporting dashboards using Python Dash and automated reporting using SMTP.
- Optimized PostgreSQL and MySQL databases through indexing and query optimization, improving execution performance by 40%.
- Developed web scraping pipelines using Selenium, BeautifulSoup, Pandas, and PySpark.
- Collaborated with cross-functional teams to define technical specifications for Mistral-based Generative AI chatbot solutions.

### Employee Management & Background Verification System

2026

*FastAPI, Python, SQLAlchemy, SQLite, LangChain, LangGraph, OpenAI, ChromaDB, SMTP*

- Developed an Employee Management System using FastAPI, SQLAlchemy, and SQLite to manage employee records, departments, and reporting hierarchy.
- Designed self-referencing manager-employee relationships using SQLAlchemy ORM.
- Implemented automated background verification workflows with SMTP email notifications.
- Built a local Retrieval-Augmented Generation (RAG) assistant using LangChain, LangGraph, OpenAI, and ChromaDB.
- Developed PDF ingestion pipelines using PyPDFLoader, document chunking, embedding generation, and semantic vector search.
- Implemented similarity search to retrieve relevant document chunks and generate accurate LLM responses.
- Ensured complete data privacy by running embedding models and LLM inference locally using OpenAI.
- Developed REST APIs for employee management, document upload, semantic search, and chatbot interactions.
- Implemented SMTP email service with automatic fallback to a local SMTP server.

**PYTHON DEVELOPER**

Sep 2021 – July 2022

**Adithri Technologies Private Limited, Hyderabad, India**

- Developed end-to-end Machine Learning pipelines using Python, Scikit-learn, Azure Anomaly Detection, NumPy, Pandas, and SciPy.
- Built scalable ETL pipelines using PySpark and Hive integrated with HDFS, Amazon S3, Kafka, and Cassandra.
- Developed RESTful microservices using Flask and Django connected with Cosmos DB, Redshift, and Azure SQL Data Warehouse.
- Designed and optimized Oracle PL/SQL, SQL Server, MySQL, and DynamoDB databases.
- Developed Angular 12 TypeScript web applications integrated with backend REST APIs.
- Managed AWS and Azure cloud infrastructure including CloudWatch, AWS Lambda, Azure DevOps, and GitHub CI/CD.

**PROJECTS**

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**SparkleClean – Full-Stack Service Booking Platform** 2025

*React, TypeScript, Tailwind CSS, Spring Boot, MySQL*

- Built a full-stack booking platform with customer and admin portals using React and TypeScript with role-based authentication.
- Designed multi-step booking workflows with real-time status tracking and CRUD operations.
- Developed backend services using Java 17, Spring Boot, and Spring Data JPA with MySQL.
- Deployed the frontend on Vercel and backend on Railway with production-grade configuration.

**Superior Limousine LLC – Corporate Website** 2025

*TypeScript, Vite, Vercel, Cloudinary*

- Built a fully responsive corporate website with a mobile-first and accessibility-focused design.
- Implemented an SEO-optimized structure to improve search engine visibility and website performance.
- Integrated Framer Motion animations and Cloudinary for optimized media delivery.
- Deployed the application on Vercel with custom DNS configuration.

**Case Notes System – NJIT** 2024

*NestJS, TypeScript, Prisma, PlanetScale, AWS Lambda*

- Developed a hospital patient case document summarization system.
- Built scalable APIs using NestJS and Prisma ORM.
- Integrated AWS Lambda for serverless processing.
- Tested APIs using Postman.

**Student Performance Prediction – NJIT** 2024

*Python, Scikit-learn, Pandas, NumPy, Matplotlib*

- Built machine learning models using Scikit-learn to predict student performance.
- Performed data preprocessing and visualization using Pandas, NumPy, and Matplotlib.

**EDUCATION**

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**Master of Science, Computer Science** May 2024

New Jersey Institute of Technology (NJIT), Newark, NJ

**Bachelor of Engineering, Computer Science** Aug 2021

Siddhartha Institute of Technology (SITS), Hyderabad, India

**CERTIFICATIONS**

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- AWS Certified Cloud Practitioner
- Generative AI Foundations
- Data Structures and Algorithms with Python